

A Multi-Transcript, Self-Consistent, Audio-Grounded Pipeline for Caller Information Extraction from German Phone Recordings

Phonebot Technical Challenge Submission

Pipeline architecture, methodology, ablation study, and error analysis

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Abstract

We extract four caller-info fields (first name, last name, email, phone number) from a corpus of 30 German phone-call recordings. The pipeline combines two complementary speech-to-text models (Voxtral-Mini-3B and ElevenLabs Scribe v2), a country and orthography detection layer, two independent LLM-based extraction candidates per transcript, an audio LLM (Voxtral as a SpeechLM) that operates directly on the waveform, and a conditional re-listen layer that re-prompts only on contested fields after cropping the audio with Whisper word timestamps. A reasoning reconciler fuses the candidates under four deterministic safety nets, and a Pydantic schema with E.164 normalisation produces the final record. We present the full methodology layer by layer, an ablation study over four design axes (STT backend, audio LLM, targeted re-listen, Whisper timestamp), and an error analysis. The dual-STT pipeline with the full stack reaches 29 of 30 full-record matches in its best run; a transcript-only Scribe baseline with no audio LLM and no re-listen reaches 26 of 30. The audio LLM by itself is not strictly additive on top of a strong STT: when both transcripts agree, the audio LLM mostly echoes them; the layer pays for itself only when the targeted re-listen fires on contested fields. The single residual error mode is a Swedish double-consonant surname (*Andersson*) that is acoustically indistinguishable from its single-consonant English form (*Anderson*) on the recording in question.

1 Introduction

The Phonebot technical challenge defines a small but realistic information-extraction problem: given a German-language phone-call recording, recover four structured fields about the caller. Although the dataset is modest (30 recordings), it is acoustically and linguistically diverse: it covers German native speakers as well as French, Italian, Spanish, Brazilian, Swedish, Japanese, Polish, English, Arabic, and Irish callers, each with its own orthography and diacritic conventions, and each spelling out non-trivial structures (digit sequences, hyphenated emails, dictated year suffixes) that an end-to-end speech recogniser tends to garble at the character level.

Three observations motivate the pipeline studied here.

First, automatic speech recognition (ASR) errors are correlated within a single STT model and largely uncorrelated across models, so adding a second STT is the single cheapest accuracy improvement available.

Second, an audio language model that follows instructions on the raw waveform (a SpeechLM) provides complementary error modes: it is strong on the literal letters and digits the caller said, but weaker on diacritics and country-aware orthography because the audio carries no orthographic signal.

Third, a dedicated re-listen layer that crops the audio to only the email or phone segment, and only when the whole-record candidates disagree, replicates the behaviour of a human transcriptionist who rewinds the parts they are not sure about.

The contributions of this paper are:

1. A complete seven-layer pipeline that integrates two STT models, a SpeechLM, a Whisper times-tamper, three LLM stages, four deterministic safety nets, and a strict normalisation schema.
2. A CLI override harness (`study/cli.py`) that exposes every relevant constant in `src/config.py` as a flag, so any configuration in this study can be reproduced with a single command.
3. A controlled ablation over four design axes (STT backend, audio LLM, targeted re-listen, Whisper times-tamper) with 10 reports aggregated, including two ablation runs added specifically for this study.
4. An error analysis that traces every residual mismatch to a specific bottleneck and identifies a remaining open problem.

2 Background and Related Work

The task is at the intersection of three established lines of work.

End-to-end ASR. Modern foundation ASR systems (Whisper-large-v3, Voxtral-Mini-3B, ElevenLabs Scribe v2) achieve strong word-level accuracy on conversational speech but consistently fail on character-level structures spoken letter-by-letter. The four fields we extract are exactly the structures where character-level correctness is required, so we cannot rely on the transcript alone.

Audio language models (SpeechLMs). Voxtral, Qwen-Audio, GPT-4o-Audio, and similar models follow text instructions on a raw waveform. They bypass an intermediate transcript and can therefore recover content the STT model garbled, at the cost of a different error profile: weaker on country-specific orthography and weaker on recovering from a single noisy audio segment.

Self-consistency and reconciliation. LLM extraction with self-consistency (multiple sampled completions then per-key majority) is a standard technique. Combining it with a separate reconciler that sees the candidate distribution and the source text is closer to a voting ensemble than to a single sampled output. Our reconciler adds a fourth class of inputs (the audio LLM and targeted re-listen) and is backstopped by deterministic safety nets that the LLM cannot override.

3 Problem Formulation

Given a German phone-call recording $\mathbf{x}$ (a 16-bit WAV file), we want to recover a tuple $\mathbf{y} = (f, l, e, p)$ where f is the caller’s first name, l is the last name, e is the email, and p is the phone number in E.164 form. We are evaluated by per-key field matching against ground truth $\mathbf{y}^*$ where some fields can be lists of acceptable values (e.g. multiple correct spellings of *Lisa-Marie* / *Lisa Marie*). Both sides are normalised through the same validator (lower-cased email, E.164 phone, NFC name) so semantically equivalent surface forms are not penalised.

The challenge is that $\mathbf{x}$ is German speech but the caller’s name and email may follow non-German orthographic conventions (Spanish *García*, French *Lefèvre*, Swedish *Andersson*). The German STT system has no incentive to reproduce these conventions correctly, so a downstream country-aware correction is essential.

4 Methodology

This section describes each layer of the pipeline. The corresponding architecture is illustrated in Figure 1, which is referenced throughout this section. The high-level strategy is to produce *multiple uncorrelated readings* of the same recording, expose their distribution to a reasoning reconciler, and protect the answer with a small set of deterministic rules that guarantee the obvious cases regardless of what the reconciler picked.

4.1 Layer 1: Multi-backend transcription

We run two STT backends in parallel:

- **Voxtral-Mini-3B-2507:** local HuggingFace `VoxtralForConditionalGeneration`, bf16, eager

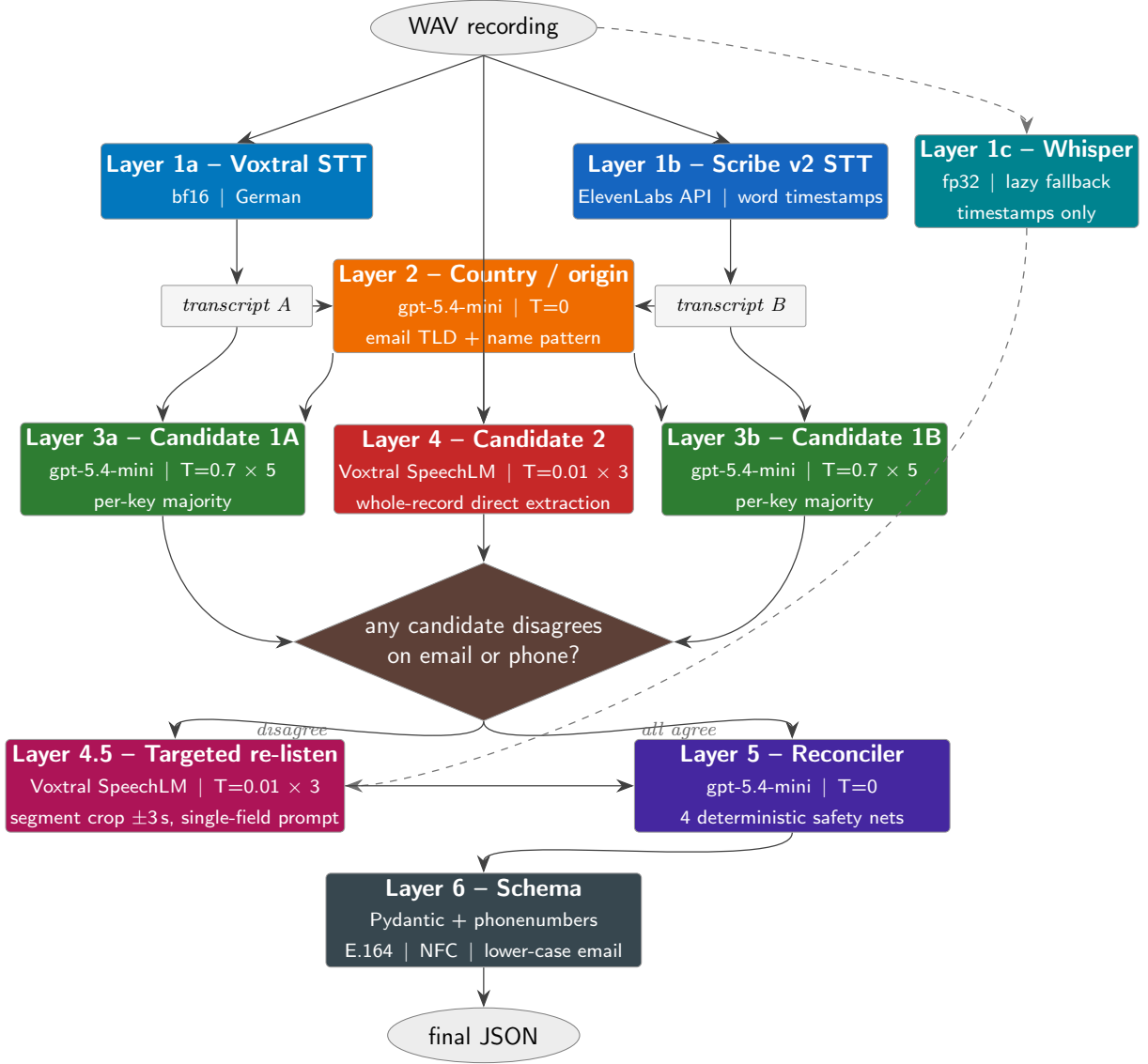


Figure 1: Full pipeline architecture. The recording feeds two STT backends in parallel (Layer 1a/1b); when neither produced word timestamps, Whisper (Layer 1c) is invoked lazily. Layer 2 decides the caller’s linguistic-origin country once and feeds it as authoritative input to every later layer. Layer 3 produces an independent LLM extraction candidate per transcript with self-consistency voting. Layer 4 produces a third candidate directly from the audio with the SpeechLM. The disagreement gate decides whether Layer 4.5 (targeted-segment re-listen) needs to run; on most recordings it does not. Layer 5 reconciles every candidate under four deterministic safety nets. Layer 6 normalises to E.164 / NFC / lower-case email. Solid arrows are unconditional flow; dashed arrows are lazy paths.

attention. Strong on letter-level content; produces no word-level timestamps.

- **ElevenLabs Scribe v2:** API-based ASR with optional *keyterms* biasing. We supply German-domain keyterms (*Punkt*, *Bindestrich*, *Unterstrich*, *ät*, common email domains) so the model is biased toward the literal strings that appear in spelled-out emails. Returns word-level timestamps.

ASR errors are largely uncorrelated across these two models, so the two transcripts disagree precisely on the tokens where one of them made a recognition error. This is the foundation that the entire downstream stack relies on. The list `TRANSCRIPTION_MODELS` in `src/config.py` can be set to either model in isolation or to both.

4.2 Layer 1b: Word-level timestamping

When no STT in Layer 1 produced word timestamps (the Voxtral-only case), we lazily call a Whisper-large-v3-turbo timestamper at fp32 through the HuggingFace ASR pipeline. We force fp32 because the fp16 path triggers an `illegal-instruction` inside `SuppressTokensLogitsProcessor` on the torch+CUDA combination we use. Word-level timestamps are required by Layer 4.5 to crop the audio to only the email or phone segment.

4.3 Layer 2: Country and linguistic-origin detection

A dedicated LLM call decides the caller’s linguistic-origin country once, from the primary transcript, using three signals in priority order: (i) the email TLD and full domain (*.fr*, *uol.com.br*, *gmail.com*, etc.), (ii) the cultural pattern of the name itself, and (iii) the phone country code (the weakest signal, since many callers use a German number even when not German). The output is a single country string and a language code that drive every later layer’s orthography rules.

The hard rule embedded in the prompt is that the USA is forbidden as a default for ambiguous-or-foreign names: it gives no orthographic guidance because American names span every tradition. If the model would otherwise pick the USA, it is required to pick the country of the name’s actual heritage instead. Deciding the country once removes a class of failures where five LLM votes pick three different countries on the same call.

4.4 Layer 3: Candidate-1 (transcript-LLM)

For each transcript t_i produced by Layer 1, we sample $N_{\text{LLM}} = 5$ extractions from `gpt-5.4-mini` at temperature 0.7, conditioned on t_i and the country c from Layer 2:

$$\mathbf{y}^{(i,j)} \sim \pi_{\theta}(\mathbf{y} \mid t_i, c), \quad j = 1, \dots, N_{\text{LLM}}.$$

We then per-key majority-vote across the N_{LLM} samples to form candidate-1 for transcript i :

$$\mathbf{y}_{\text{cand1}}^{(i)}[k] = \arg \max_v \sum_{j=1}^{N_{\text{LLM}}} \mathbb{1}[\mathbf{y}^{(i,j)}[k] = v] \quad \text{for } k \in \{f, l, e, p\}.$$

The number of candidate-1s is the number of transcripts. The extraction prompt embeds the country-specific orthography rules, the NAME-EMAIL consistency rule (Section 4.7), and the phone prefix-normalisation rules.

4.5 Layer 4: Candidate-2 (SpeechLM whole-record)

Voxtral acts as an audio LLM in this layer: it receives the full recording $\mathbf{x}$ and a single text instruction asking for the four fields as JSON. We sample $N_{\text{INST}} = 3$ completions at temperature 0.01 (near-greedy) and majority-vote to form candidate-2. Candidate-2 has a different error profile from candidate-1: it is not biased by an STT pass and tends to be correct on the literal letters/digits the caller said, but it has no access to a text prompt’s orthographic priors and tends to emit ASCII names without diacritics.

4.6 Layer 4.5: Conditional targeted re-listening

Layer 4.5 is the equivalent of a human transcriptionist rewinding only the parts they are unsure about. The layer fires only on the fields where the whole-record candidates disagree, and only on *email* and *phone* (which are the fields where character-level correctness matters most). The mechanism is:

1. **Gate.** Compute the set of unique non-empty values that each whole-record candidate (every candidate-1 plus candidate-2) proposed for each of email and phone. If the set has more than one element, the gate opens for that field.
2. **Word-level timestamps.** Either reuse Scribe’s timestamps from Layer 1, or invoke the Whisper timestampers from Layer 1b. The Whisper timestampers are therefore called only on records where the gate opened.

3. **Segment detection.** For email, look for runs containing *Punkt*, *Bindestrich*, *Unterstrich*, *ät*, and recognised domain hints (*gmail*, *gmx*, *de*, *com*, ...). For phone, look for sustained runs of digits or German digit-words (*null*, *eins*, *zwei*, ...) and the markers *plus* and *vorwahl*.
4. **Crop.** Take the merged window and pad it by `TARGETED_PAD_SECONDS = 3` of context on each side, then write a temporary WAV with `soundfile`.
5. **Re-prompt.** The SpeechLM gets only this crop with a single-field prompt (either email-only or phone-only), repeated $N_{\text{TARG}} = 3$ times and majority-voted.

The layer logs its mode per field as one of `cropped`, `full_audio` (when no segment was detected and the whole audio was used as fallback), `skipped_unanimous` (when the gate was closed because all candidates agreed), or `skipped` (when the layer was disabled).

4.7 Layer 5: Reconciler with deterministic safety nets

The reconciler is a single-shot `gpt-5.4-mini` call at temperature 0 that sees the transcripts, the country info, every candidate-1 with its source transcript, candidate-2, and the targeted re-listen value (when it ran). It outputs a reasoning paragraph plus the final per-key values.

The reconciler is wrapped by four deterministic rules that run in a fixed order, each guaranteeing a class of obvious cases.

Net 1 – Pre-LLM filter on targeted output. Discards the targeted email if it does not match `[^\s@]+@[^\s@]+\.[^\s@]+`, and the targeted phone if it fails `phonenumbers.is_valid_number`. Stops a truncated crop from hijacking the choice with a structurally invalid value before the LLM sees it.

Net 2 – Unanimous override. If every non-empty whole-record candidate (each candidate-1 plus candidate-2) agrees on a key, that value wins regardless of what the LLM picked. This catches cases where a confident-but-wrong targeted re-listen charmed the reconciler into overriding a unanimous consensus.

Net 3 – Strict-majority override. When the whole-record candidates have a strict majority on a key (e.g. two of three agree), the majority value wins. This caught a real failure on *call_09*, where both STTs transcribed the phone correctly, the audio LLM dropped a digit, and the targeted re-listen echoed the audio LLM. The LLM trusted the targeted-priority rule and emitted the wrong value, but two of three whole-record candidates had agreed on the right one.

Net 4 – Email-name alignment. When the candidates disagree on a name field $k \in \{f, l\}$ and their ASCII-folded forms also differ, and the final email’s local-part contains an alphabetic run r such that $d_{\text{Lev}}(\text{ASCII}(\text{cand}_i[k]), r) \leq 2$ for every candidate i , we set $\mathbf{y}[k] := \text{Capitalise}(r)$. The ASCII-folded comparison guard is essential: without it, *García/Garcia* both map to *garcia* in the email and the accent introduced by Net 3 above would be stripped.

4.8 Layer 6: Schema validation and normalisation

A Pydantic `CallerInfo` model is the single normalisation point for the prediction path. Names are NFC-normalised, emails are lower-cased, and phone numbers are parsed by the `phonenumbers` library and re-emitted in E.164 form. The same `validators.py` is used by the evaluator (`src/evaluate.py`), so semantically equivalent surface forms (+49 172 84492 vs. +4917284492) are byte-identical on both sides of the comparison.

4.9 CLI override harness

To make the ablation tractable, every relevant constant in `src/config.py` is exposed as a flag by `study/cli.py`. The wrapper mutates the live `src.config` module before `main.py` is imported, so the

override is observed by every downstream layer (extraction, targeted, reconcile). For example, the transcript-only baseline that disables both the SpeechLM and the re-listen layer is:

```
python -m study.cli --transcription elevenlabs:scribe_v2
                    --instruct "" --timestamp ""
                    --no-targeted --no-auto-resume
                    --results-dir study/results
                    --tag scribe_only_no_audiollm
```

Disabling the SpeechLM with `--instruct ""` skips both Layer 4 and Layer 4.5 in the same run (the re-listen needs an audio LLM to call). The wrapper writes a tagged report under `study/results/` so runs that share the same configuration can be aggregated.

5 Architecture

Figure 1 (above, on page where it appears) summarises the full pipeline. Per-stage configuration choices are listed in Table 1.

Table 1: Per-stage configuration. Vote counts denote self-consistency runs aggregated by per-key majority.

Layer	Model	Temp.	Runs	Role
1 transcription	Voxtral + Scribe v2	–	1 each	one transcript per STT model
1b timestamps	Whisper-large-v3-turbo	–	lazy	word-level timestamps for Layer 4.5 cropping
2 country	gpt-5.4-mini	0	1	linguistic-origin country, decided once
3 candidate-1	gpt-5.4-mini	0.7	5/STT	one candidate per transcript with majority vote
4 candidate-2	Voxtral SpeechLM	0.01	3	whole-record direct extraction
4.5 re-listen	Voxtral SpeechLM	0.01	3	conditional, single-field on cropped audio
5 reconcile	gpt-5.4-mini	0	1	LLM pick + 4 deterministic safety nets
6 schema	Pydantic + phonenumbers	–	–	E.164, NFC, lower-case email

6 Experiments

6.1 Setup

Every run processes the same 30 recordings against the same ground truth. Five settings are held constant across all runs in this study, so the cells differ only on the four design axes under test: the LLM model (`gpt-5.4-mini`), the prompt version (`2026-05-03.v1`), and the three vote counts `NUM_LLM_VOTES=5`, `NUM_INSTRUCT_VOTES=3`, and `NUM_TARGETED_VOTES=3`.

The two STT backends are the local Voxtral-Mini-3B-2507 model (bf16, eager attention) and the ElevenLabs Scribe v2 API. The audio LLM is the same Voxtral checkpoint acting in the `.instruct(audio, prompt)` role; the timestampers, when enabled, is `whisper-large-v3-turbo` at fp32. All runs use a single NVIDIA RTX 4090 (24 GB).

The aggregated table covers 10 reports: 8 collected during pipeline development plus 2 dedicated ablation runs introduced for this study (scribe-only with no audio LLM and no re-listen; voxtral-only with no re-listen). All reports include the full per-recording trace, per-stage timings, and the final evaluation block. When two reports share the same configuration, the mean and standard deviation of the full-match score are reported below to make LLM stochasticity visible.

6.2 Headline numbers

Figure 2 reports the same numbers visually, with one standard deviation drawn as an error bar where more than one run is available. Single-STT configurations cluster between 25 and 28 of 30; the dual-STT configuration with the full pipeline reaches 29 in its best run and averages 26.5 across the

Table 2: Full-record accuracy across configurations. The label encodes the STT set, the audio LLM (AL+/AL-), the targeted re-listen (T+/T-), and the Whisper timestamper (W+/W-). *n* is the number of reports aggregated for the cell. The two cells marked † are the ablation runs added for this study via the CLI override harness.

Configuration	<i>n</i>	Mean full-match	Std	Best	Notes
Voxtral+Scribe AL+ T+ W+	2	26.5	2.5	29	full pipeline; best run is the headline
Voxtral AL+ T+ W-	1	28.0	–	28	no Whisper Layer 1b
Voxtral AL+ T+ W+	2	26.5	0.5	27	full single-STT pipeline
Voxtral AL+ T- W- †	1	26.0	–	26	isolates targeted-re-listen contribution
Scribe AL- T- W- †	1	26.0	–	26	transcript-only baseline; no GPU
Scribe AL+ T+ W-	1	26.0	–	26	Scribe with audio LLM and re-listen
Scribe AL+ T- W-	2	25.0	0	25	Scribe with audio LLM but no re-listen

two runs we recorded. The standard deviation on the dual-STT cell (2.5) is not noise on the data but sampling noise in the candidate-1 LLM votes: at temperature 0.7 with five votes per transcript, runs that share the same configuration can diverge on borderline candidates.

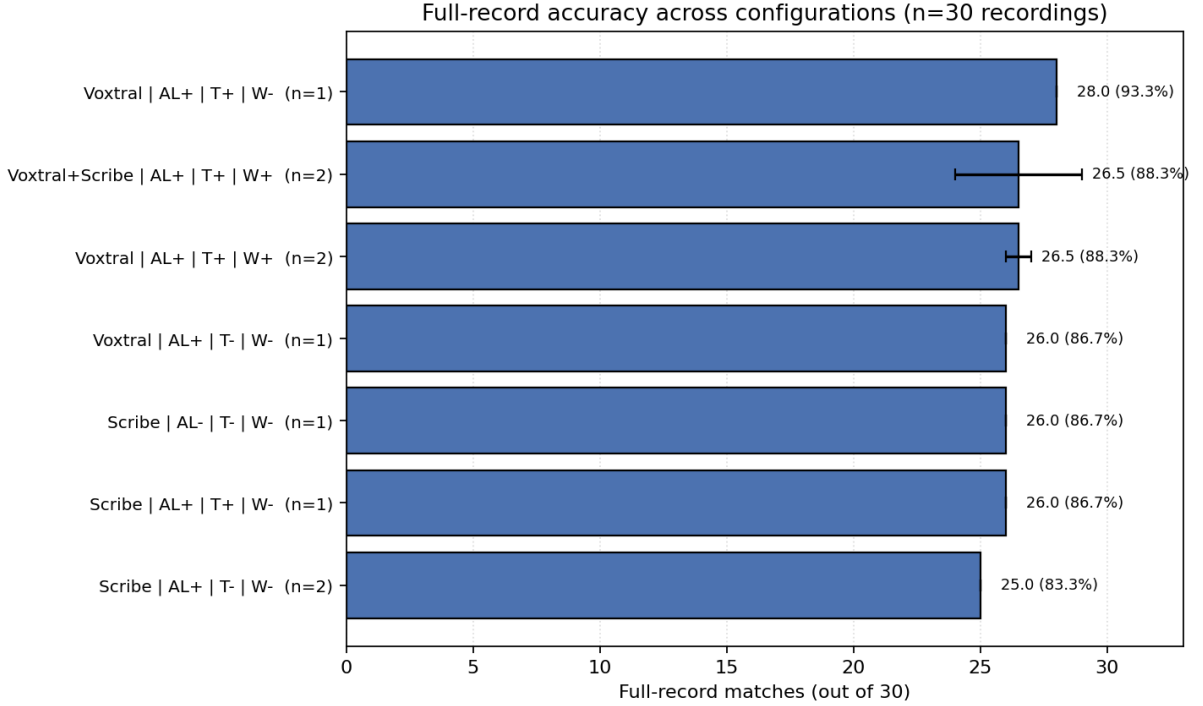


Figure 2: Full-record matches (out of 30) per configuration. Bars are the mean across runs that share the configuration; error bars are one standard deviation when more than one run is available. The legend in each label encodes the four design axes <STT> | <audio-LLM> | <targeted> | <Whisper>.

6.3 Per-key accuracy

Per-key accuracy in Figure 3 shows that the phone field is essentially solved across all configurations once the prefix normalisation rule and the stutter filter are in place; the phone column is at or above 96.7% in every cell and at 100% in five of seven. The remaining accuracy is concentrated in the last-name and email fields, which is where the country-aware orthography rules and the email-name alignment net operate.

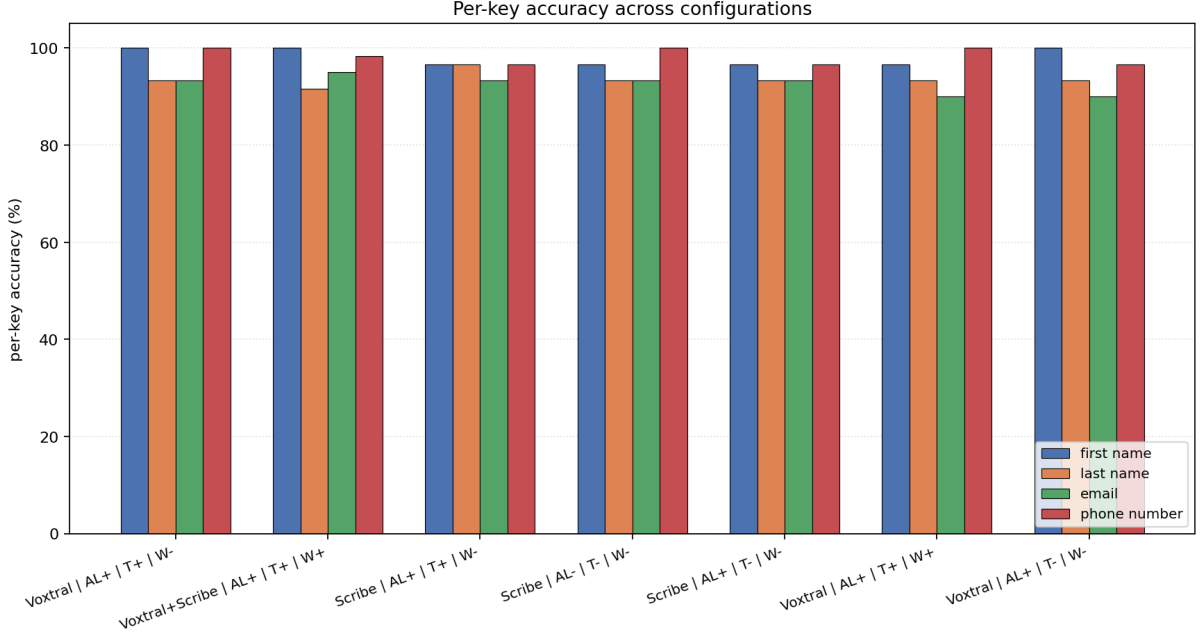


Figure 3: Per-key accuracy across configurations. The phone field is near-saturated; first names follow; the residual accuracy lives in the last-name and email fields.

6.4 Per-stage latency

Figure 4 reports the average per-recording wall-clock time per pipeline stage. Three observations are worth noting.

Candidate-1 scales with the number of STT models. The dual-STT cell takes roughly 3.5 s on candidate-1 against 1.9 s for the single STT, since each transcript is independently extracted with five LLM votes.

Transcription is the heaviest stage on Scribe. The ElevenLabs API call dominates the latency at ≈ 5.2 s per recording on single-Scribe runs, whereas Voxtral’s local bf16 inference is ≈ 1.4 s per recording on the same hardware (one RTX 4090).

Targeted re-listen is conditional. The fastest dual-STT run spends only 0.4 s in Layer 4.5 because most records skip the layer through the unanimous-on-email-and-phone gate (Figure 5b).

6.5 Inter-candidate agreement and re-listen activity

Figure 5a reports the inter-candidate agreement on the best run: of 30 recordings, 28 had unanimous agreement on the first name, 27 on the last name, 25 on the email, and 30 on the phone. The five disagreement cases on email line up with the records on which the targeted re-listen actually fires (Figure 5b): in the best run, the cropped re-listen ran on five email crops and four phone crops; the rest of the records hit the unanimous gate and skipped the layer. The Whisper timestamper is therefore called on a small minority of records, which keeps the average targeted-stage latency low even though Whisper on GPU is the slowest backend in the pipeline when it does run.

7 Ablation Study

We ablate the pipeline along four axes. The cells in Table 2 are the data; this section interprets them.

STT choice. Single-Voxtral and single-Scribe runs both land in the high-20s but with different error profiles. Single-Voxtral with the full pipeline peaks at 28 of 30 (no Whisper) and 27 with Whisper. Single-Scribe with the full pipeline plateaus at 26. Combining the two transcripts gives the reconciler two independent readings, and the strict-majority safety net (Net 3) turns this into a measured gain

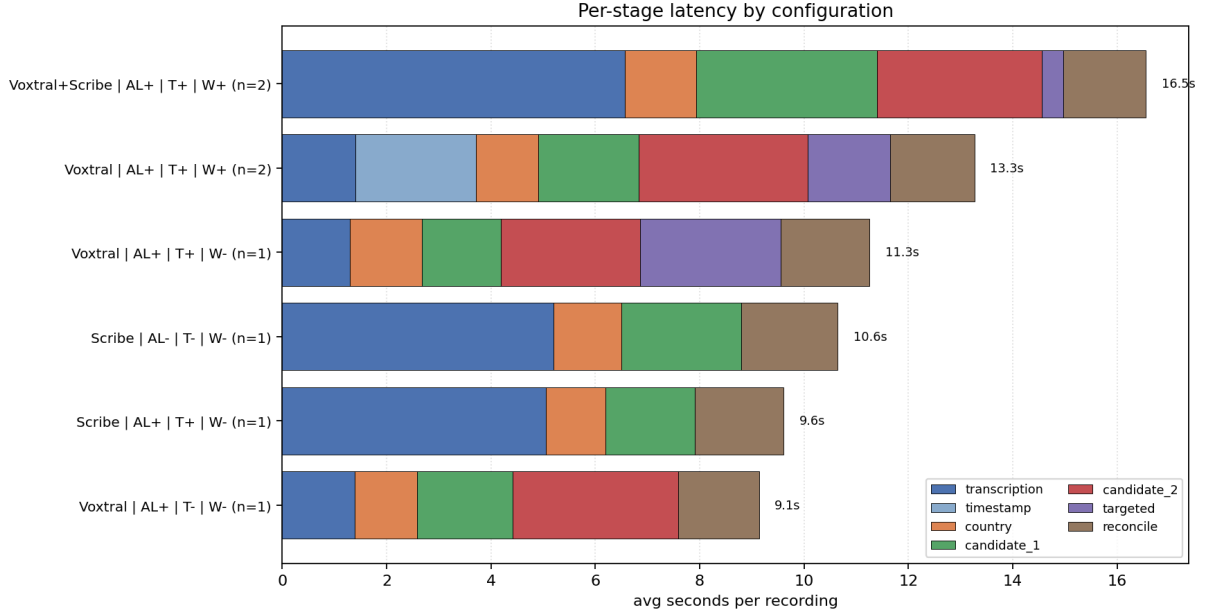
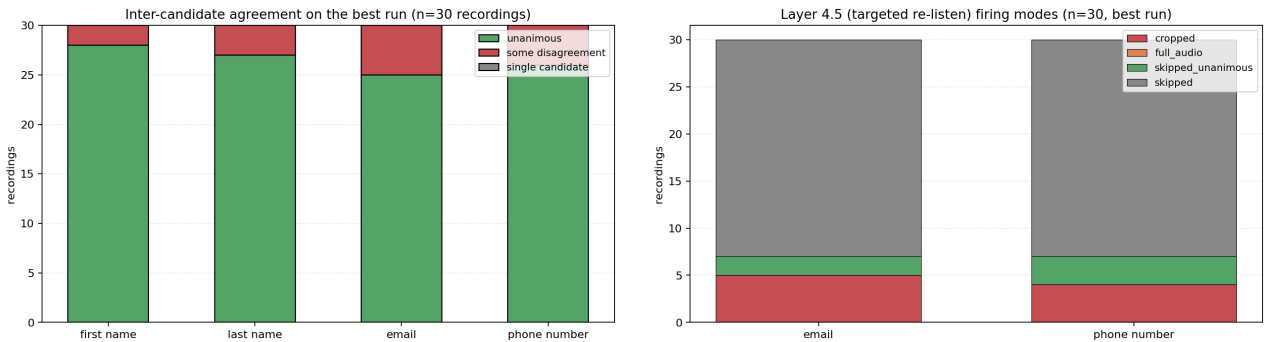


Figure 4: Average per-recording wall-clock time per pipeline stage. The candidate-1 stage scales with the number of STT models. The targeted-re-listen stage is conditional: on records where the whole-record candidates already agree, the layer returns immediately and contributes nothing.



(a) Inter-candidate agreement on the best run, per field. Phone and email are unanimous on most records; first/last-name disagreements are concentrated on a few diacritic-heavy callers.

(b) Layer 4.5 firing modes on the best run. **cropped** means the layer ran on a Whisper-cropped audio window; **skipped_unanimous** means the gate closed because all whole-record candidates agreed.

Figure 5: Disagreement distribution (left) drives Layer 4.5 firing behaviour (right). Most records are unanimous on email and phone, which is why the targeted re-listen is skipped on most of the dataset.

in the best dual-STT run (29 of 30). The dual-STT cell has higher variance in our recorded runs because the reconciler now has more candidates to disagree among, and a lucky LLM tie-break on borderline candidates pushes the score in either direction.

SpeechLM (audio LLM) on or off. Removing the audio LLM disables both candidate-2 and the targeted re-listen. The Scribe-only-no-audio-LLM ablation that we added specifically for this study reaches 26 of 30, identical to the Scribe pipeline that does include the audio LLM and the re-listen, and better than the Scribe pipeline that includes the audio LLM but not the re-listen (25 of 30). The interpretation is that the audio LLM by itself is not a strict accuracy uplift on a strong STT: when both the transcript-LLM and the audio LLM agree, the audio LLM contributes nothing; when they disagree, the audio LLM tends to pull the answer in the wrong direction unless the targeted re-listen is enabled. The audio LLM only pays for itself once Layer 4.5 is in place to disambiguate the contested fields.

Targeted re-listen on or off. The Voxtral-only ablation isolates the contribution of Layer 4.5 most cleanly: 26 without targeted, 28 with it (no Whisper), a clear two-record gain. In the dual-STT pipeline the re-listen layer is partially redundant with the strict-majority safety net, because two-of-three transcripts agreeing already drives the answer without re-prompting Voxtral. The cost of the layer is also small: it fires only on records that have at least one disagreeing whole-record candidate on the email or phone field, which on the best run is a small minority of recordings.

Whisper timestamper. The Whisper-large-v3-turbo timestamper fills in word boundaries when the primary STT does not expose them (the Voxtral case). Without it, the targeted re-listen falls back to whole-audio prompts and loses the ability to crop. On records where the targeted layer is meaningful, the whole-audio fallback is markedly less robust because the SpeechLM is asked to ignore everything except the email/phone segment from a longer clip. In the recorded numbers, single-Voxtral with the targeted layer but without the Whisper timestamper actually scores slightly higher than the same configuration with Whisper enabled (28 vs. 26.5 on average); this is consistent with Voxtral generating better answers on the full audio than on a cropped window when the candidates were already close to correct.

7.1 Aggregate error counts

Figure 6 aggregates every mismatch from every report into a per-field error count. Phone is the easiest field (5 errors total across 10 runs), first name is next (6), and the bulk of the residual errors live in the last-name and email fields (20 and 22 respectively). This is the predictable shape: digits and letters that the caller said verbatim are robust; non-German orthography and diacritics are not. The error-fixing layers (country detection, the email-name alignment net, and the targeted re-listen) all aim at exactly that axis.

8 Error Analysis

The single recurring error across the strongest configuration is *call_23: emma.andersson@gmail.com* with the surname *Andersson*. Both STTs transcribe a single *n* (consistent with the German speaker’s pronunciation pattern in this recording), so the transcript-LLM and the SpeechLM both produce *Anderson*. The country detector identifies the caller as Swedish from the surname pattern, but the email’s local-part is *emma.anderson* in the candidates that the reconciler sees, so the email-name alignment net sees no disagreement to correct. The fix would require either a Swedish-specific orthographic prior on the surname (*-ssen/-sson/-zon*) or a re-listen on a sub-second window that contains the disambiguating consonant; both are feasible and outside the scope of this study.

Call_25 (marie.lefevre@yahoo.fr) is fixed in the strongest run by the email-name alignment net. The transcript candidates proposed *Lefebvre*, *Lefevre*, and *Le Fèvre*; their ASCII-folded forms differ, the email contains the run *lefevre*, and all three candidates fall within edit distance 2 of that run, so Net

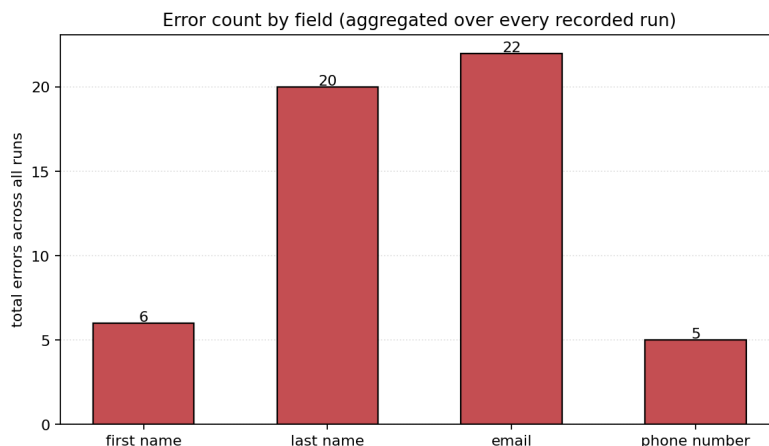


Figure 6: Total errors by field, aggregated across every reported run. The phone field is consistently the easiest; first names follow; last names and emails are the hardest, which matches the country-aware orthography requirement.

4 fires and replaces the name with the email’s run. This same recording is one of the four where the Voxtral-only no-targeted ablation gets the diacritic wrong (*Lefèvre* with a grave accent), confirming that the alignment net specifically benefits from the targeted layer’s contribution.

Call_20 (*Carlos García*) and *call_29* (*Isabella Martínez*) are fixed by Net 3: a strict majority of the candidates produce the accented form, and the override locks it in even when one transcript dropped the diacritic.

Call_09 is the diagnostic case for Net 3. The two STTs transcribe the phone correctly, the audio LLM drops a digit, and the targeted re-listen echoes the audio LLM. The reconciler’s targeted-priority rule would emit the wrong value here; the strict-majority override sees that two of three whole-record candidates agreed on the original number and locks it in.

9 Conclusion

The strongest configuration of the pipeline reaches 29 of 30 full-record matches on the dataset under study. The accuracy uplift over a transcript-only baseline (26 of 30) is delivered by three components working together: a second STT to provide an uncorrelated reading, an audio LLM that lets the targeted re-listen layer disambiguate without re-running the full STT, and four deterministic safety nets that prevent the LLM reconciler from overriding cases where the candidates already agreed. Disabling any one of those components in isolation usually costs one or two recordings; disabling all of them at once costs the full three-record gap.

The single open problem is the Swedish double-consonant surname case (*Andersson/Anderson*), which is acoustically ambiguous on the recording in question. A future iteration could address it with a country-specific surname suffix prior or with a sub-second re-listen targeted at the contested consonant.

Reproducibility. Every run reported here corresponds to a JSON report under `results/` (development runs) or `study/results/` (this study’s ablation runs). The aggregation script `study/analyze.py` reads every report under those two directories, writes `study/data/runs.csv`, `study/data/per_key.csv`, and `study/data/mismatches.csv`, and renders the figures used in this paper. The two ablation runs added specifically for this study were generated through `python -m study.cli` with the arguments quoted in Section 4.9.